PARABOLA BASS Z

Manual for v0.4: a 3-band parallel bass processor with a NAM capture per band



The Foil skin with captures loaded. Settings (the gear) has four more skins.

The signal path



Mid and high go through the cab by default; low skips it (each band's ADV page has a CAB switch). A capture that already includes a cab skips it automatically.

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1 Install and first sound

1. Run **Parabola-Bass-Z-v...-windows-x64.exe**. Windows SmartScreen may warn the first time: *More info* → *Run anyway*. It installs the VST3 plugin, the standalone app, and makes `Documents\Parabola Bass Z\NAMs` and `\Cabs` for your files.
2. Download captures and a cab IR (TONE3000 has plenty; none ship with Parabola) into those folders, or keep them anywhere and add that folder in **MY GEAR → FOLDERS**.
3. **In Reaper:** Options → Preferences → Plug-ins → VST → *Re-scan*, then add Parabola to your bass DI track. It replaces your amp and cab. **Standalone:** Start menu → Parabola Bass Z, gear → ASIO, your interface, your bass's input, **Hear yourself** on.
4. Pick a starter preset from the **PRESET** bar (it finds its captures by file name), or open **MY GEAR** and click captures into bands. **LOAD** and drag-and-drop work too. Balance with the Level knobs.

The ADV side of a band



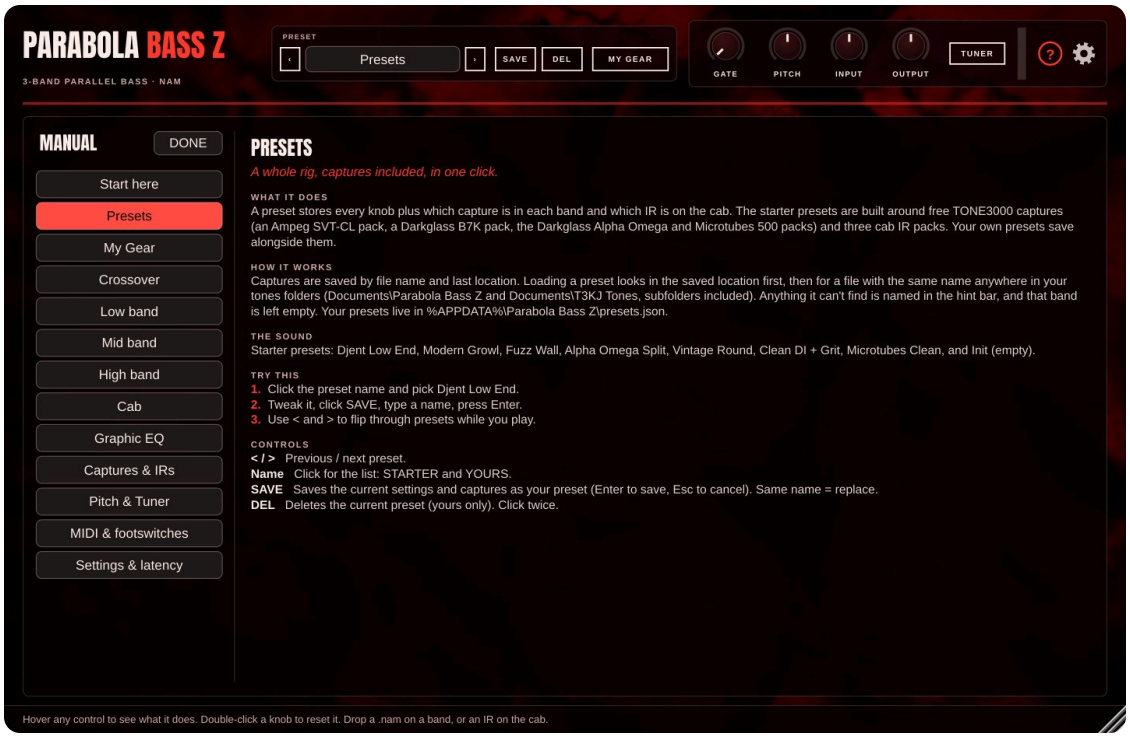
ADV flips each band to Bass / Mid / Treble and its CAB switch (Ember skin). The note next to CAB says whether the band goes through the IR.

Moves that work everywhere

DO THIS	TO
Drag a knob up / down	Turn it. While dragging, its label shows the value.
Double-click a knob, fader or crossover handle	Reset it.
Hover anything	Read what it does in the hint bar at the bottom.
Drag a .nam onto a band, a .wav onto the cab	Load it.
Drag the window's bottom-right corner	Resize, 60% to 200%. It remembers.

R Reference

The same text as the manual inside the plugin: click **?** in the header, pick a topic on the left.



2 Start here *Three amps on one bass, each on its own slice of the range.*

What it does. Parabola splits your bass into three bands (low, mid, high) and gives each band its own amp capture (a NAM file you load). The low band stays clean and compressed so the bottom never gets mushy; the mid band growls; the high band adds grit and fizz. Then a cab IR goes on the mid and high bands, a graphic EQ shapes the total, and it all comes back out as one signal.

How it works. This is the trick behind parallel bass rigs like the Parallax X: distortion on the lows turns to mud, so the lows are kept clean and the drive happens above them. Parabola does the split with phase-aligned crossovers, so with nothing loaded and every band at 0 dB, your signal comes out unchanged.

The sound. Modern metal and djent bass: a tight, round low end that holds up the mix, with a snarling, articulate top on it, and every part adjustable on its own.

Try this.

1. Put Parabola on your bass track (DI, no amp before it). It replaces your amp and cab.
2. Pick a starter preset (the PRESET bar): it finds its captures by file name in your tones folders.
3. Or open MY GEAR and click captures to load them into bands, and an IR into the cab.
4. Hover anything: the bar at the bottom says what it does. Drop .nam files right onto the bands.

CONTROL	WHAT IT DOES
Hush	The de-noiser on your DI, first in the chain (zero latency). Click HUSH in the header.
Presets	< and > step through them, click the name for the list, SAVE stores yours, DEL deletes yours.
My Gear	Every capture and IR in your tones folders, one click to load.
Gate	Silences noise between notes, on your DI before everything. Fully down is off.
Noise	A second gate after the mid and high bands, following your DI: shuts the drive bands' hiss as notes fade while the low band rings on. Fully down is off.
Pitch	Transposes everything ±12 semitones before the bands.
Input / Output	Level into Parabola and out of it, ±24 dB. The meter turns red when the output clips.
Tuner	Opens the tuner and mutes the output.
?	This manual.
Gear	Settings: the skin, and in the standalone app your audio interface and latency.

3 Presets *A whole rig, captures included, in one click.*

What it does. A preset stores every knob plus which capture is in each band and which IR is on the cab. The starter presets are built around free TONE3000 captures (an Ampeg SVT-CL pack, a Darkglass B7K pack, the Darkglass Alpha Omega and Microtubes 500 packs) and three cab IR packs. Your own presets save alongside them.

How it works. Captures are saved by file name and last location. Loading a preset looks in the saved location first, then for a file with the same name anywhere in your tones folders (Documents\Parabola Bass Z, Documents\T3KJ Tones and any folder you added in My Gear, subfolders included). Anything it can't find is named in the hint bar, and that band is left empty. Your presets live in %APPDATA%\Parabola Bass Z\presets.json.

The sound. Starter presets: Djent Low End, Modern Growl, Fuzz Wall, Alpha Omega Split, Vintage Round, Clean DI + Grit, Microtubes Clean, and Init (empty).

Try this.

1. Click the preset name and pick Djent Low End.
2. Tweak it, click SAVE, type a name, press Enter.
3. Use < and > to flip through presets while you play.

CONTROL	WHAT IT DOES
< / >	Previous / next preset.
Name	Click for the list: STARTER and YOURS.
SAVE	Saves the current settings and captures as your preset (Enter to save, Esc to cancel). Same name = replace.
DEL	Deletes the current preset (yours only). Click twice.

4 My Gear *Your captures and IRs, one click away.*

What it does. A list of every capture (.nam) and IR (.wav, .aiff, .flac) in your tones folders, grouped by the folder they're in, with search. Keep your captures wherever you like: add that folder once with FOLDERS.

How it works. It reads Documents\Parabola Bass Z, Documents\T3KJ Tones (T3KJ's My Gear folder) and any folder you add with FOLDERS (subfolders included), so nothing needs a second copy. Click a capture, then pick LOW, MID or HIGH. Click an IR and it goes straight onto the cab.

The sound. Quick A/B-ing: keep My Gear open and swap the mid band's capture until the growl sits right.

Try this.

1. Click MY GEAR, then FOLDERS > Add a folder, and pick the folder you unzip captures into (C:\Users\you\NAM Tones (Bass), say).
2. Type 'b7k' in the search box, click B7K Djent, pick MID.
3. Added files? Click RESCAN.

CONTROL	WHAT IT DOES
Search	Filters by name and folder.
FOLDERS	Add a folder to look in, or remove one you added. Presets search them too.
RESCAN	Reads the folders again.
OPEN FOLDER	Opens Documents\Parabola Bass Z (and makes NAMs and Cabs inside it).

5 Crossover *Where the low band ends and the drive begins.*

What it does. The display shows the three bands as colored curves, including each band's level and EQ. The two white handles are the split points: Low / Mid (default 180 Hz) and Mid / High (default 1.6 kHz).

How it works. Each split is a Linkwitz-Riley 24 dB/octave filter pair, the standard for speaker crossovers because the two sides add back up flat. The low band also passes through a matching all-pass, so all three bands stay in phase with each other.

The sound. A higher Low / Mid split keeps more of the bass clean (rounder, more polite). A lower one lets the drive reach further down (meaner, but watch for mud). The Mid / High split decides how much of the growl is 'mid' versus 'fizz'.

Try this.

1. Start at 180 Hz / 1.6 kHz. For 5-string or drop tunings try 150 Hz.
2. Mute the mid and high bands (their power lights) to hear just the low band, then sweep Low / Mid.
3. Double-click a handle to put it back to its default.

CONTROL	WHAT IT DOES
Low / Mid handle	Drag left or right: 60 to 400 Hz.
Mid / High handle	Drag left or right: 600 Hz to 5 kHz.
Legend	Each band's level, or OFF when its power light is off.

6 Low band *The clean, compressed foundation.*

What it does. Everything below the Low / Mid split. It can run a capture (a clean amp is the classic choice) or nothing at all for a pure DI low end, then goes through its own compressor.

How it works. Drive sets the level into the capture. Comp is a one-knob compressor: turning it up lowers the threshold and raises the ratio, with a little makeup gain, so the lows sit at an even level whatever you play. By default the low band skips the cab IR, the Parallax way: a cab would only smear the fundamental.

The sound. The weight and the pitch of every note: solid, even, never fuzzy.

Try this.

1. Leave it empty, or load a clean amp (an Ampeg SVT clean, say). Comp around 40%.
2. More Comp for picked, aggressive parts; less for fingerstyle dynamics.
3. ADV: a little Bass boost here instead of in the graphic EQ keeps the drive bands tight.

CONTROL	WHAT IT DOES
Power light	Mutes the band.
LOAD / ×	Load a .nam capture (or drop one on the band), or remove it.
Drive	Level into the capture, -12 to +24 dB.
Comp	Compression amount, 0 to 100%.
Level	The band's volume in the mix.
ADV: Bass / Mid / Treble	Shelf at 100 Hz, bell at 650 Hz, shelf at 3 kHz, ±12 dB.
ADV: CAB	Send this band through the cab IR. Off by default for low.

7 Mid band *The growl.*

What it does. Between the two splits: where a bass's character and most of its distortion live. Load a drive capture here (a Darkglass-style pedal or a driven amp).

How it works. Drive pushes the band harder into its capture. Each capture's output is level-matched from its loudness data, so swapping captures doesn't jump the volume. Tone is a low-pass after the capture that rounds off the top of the distortion. Through the cab IR by default.

The sound. Growl, bark and definition: what makes a riff audible in a dense mix.

Try this.

1. Load B7K Djent or B7K Hard Rock. Drive 0 dB to start, then up to taste.
2. Tone around 5 kHz keeps the growl without fizz; open it up if the high band is off.
3. Level a couple of dB under the low band is a good starting balance.

CONTROL	WHAT IT DOES
Power light	Mutes the band.
LOAD / ×	Load a .nam capture (or drop one on the band), or remove it.
PRE / POST	PRE (default) splits first: the capture only hears this band's slice. POST drives first: the capture hears everything above the low split, and the slice is cut out after it. POST sounds thicker and a bit quieter.
Drive	How hard the capture is pushed, -12 to +24 dB.
Tone	Low-pass after the capture, 1 to 20 kHz (fully up is off).
Level	The band's volume in the mix.
ADV: Bass / Mid / Treble	Shelf at 100 Hz, bell at 650 Hz, shelf at 3 kHz, ±12 dB.
ADV: CAB	Send this band through the cab IR. On by default.

8 High band *Grit, fizz and pick attack.*

What it does. Everything above the Mid / High split: string noise, pick attack and the fizzy edge of a distorted bass. A heavier drive capture usually goes here.

How it works. Same as the mid band: Drive into the capture, level-matched output, a Tone low-pass, Bass / Mid / Treble, then the cab IR. Because only the top of the range reaches this capture, you can drive it very hard without the low end turning to mush.

The sound. The aggressive top end metal bass is known for. A little goes a long way; it's usually the quietest band.

Try this.

1. Load B7K Heavy or B7K Distortion. Level around -6 dB, Mid / High split around 1.2 to 1.6 kHz (higher than that and a bass gives the band almost nothing but string noise).
2. Hissy? Turn up NOISE in the header (start around -55 dB), lower this band's Drive, and bring Tone down to 8 kHz.
3. Try POST with Drive around -6 dB for a thicker grind; raise Level to match.

CONTROL	WHAT IT DOES
Power light	Mutes the band.
LOAD / ×	Load a .nam capture (or drop one on the band), or remove it.
PRE / POST	PRE (default) splits first: the capture only hears this band's slice. POST drives first: the capture hears everything above the low split, and the slice is cut out after it. POST sounds thicker and a bit quieter.
Drive	How hard the capture is pushed, -12 to +24 dB.
Tone	Low-pass after the capture, 1 to 20 kHz (fully up is off).
Level	The band's volume in the mix.
ADV: Bass / Mid / Treble	Shelf at 100 Hz, bell at 650 Hz, shelf at 3 kHz, ±12 dB.
ADV: CAB	Send this band through the cab IR. On by default.

9 Cab *A speaker cabinet for the drive bands.*

What it does. Load a cab impulse response (a .wav) and it's applied to every band whose CAB switch is on: mid and high by default. The top line says which bands go through it.

How it works. The IR is convolved (zero added latency) with the sum of the cab bands. A capture whose file says it already includes a cab (a 'full rig' capture, like the Alpha Omega pack) skips the IR automatically, so it isn't doubled up.

The sound. The cab takes the raw fizz off the distortion and gives it the shape of a real speaker. 8×10s are big and round, 4×10s tighter and punchier, 2×15s warm and vintage.

Try this.

1. Try an Ampeg 8×10 with a D6 mic, then an Eden 4×10, and compare.
2. CAB ON off for a DI-style tone (with a guitar-style cab sim later in the chain, say).
3. Drag a .wav straight onto the cab panel.

CONTROL	WHAT IT DOES
LOAD / ×	Load an IR (.wav, .aiff, .flac) or remove it.
CAB ON	Runs the cab bands through the IR. Off bypasses it for every band.

10 Graphic EQ *Final shaping, after the bands are back together.*

What it does. Six faders, ±12 dB, at 80 Hz, 250 Hz, 500 Hz, 1 kHz, 2.5 kHz and 6 kHz, on the mixed signal.

How it works. Six bell filters in a row. A fader at 0 dB is switched out completely.

The sound. 80 Hz is the thump, 250 Hz the boom (cut it for clarity), 500 Hz the wood, 1 kHz the honk and clank, 2.5 kHz the growl and attack, 6 kHz the sizzle.

Try this.

1. Cut 250 Hz by 2 to 4 dB to clean up mud.
2. Boost 1 kHz or 2.5 kHz a little to cut through guitars.
3. Double-click a fader to reset it.

CONTROL	WHAT IT DOES
Faders	Drag up to boost, down to cut. Double-click resets.

11 Captures & IRs *Where the sounds come from.*

What it does. Parabola doesn't ship with any captures or IRs: you load your own .nam files (from TONE3000, say) and cab IRs. Their licenses usually don't allow passing the files around, so each person downloads their own.

How it works. Parabola remembers each band's file and the IR by their location on your computer, so a project or a saved session finds them again as long as they stay put. Put them in Documents\Parabola Bass Z\NAMs and \Cabs (the installer makes those folders) and the LOAD buttons open there.

The sound. Good starting points: a clean Ampeg SVT for low, a Darkglass B7K pack for mid and high, an Ampeg 8×10 or Eden 4×10 IR.

Try this.

1. Download captures, drop them into Documents\Parabola Bass Z\NAMs (or keep them in T3KJ Tones).
2. Open MY GEAR and click one to load it, or drag a .nam onto a band.
3. The starter presets look for specific packs by file name: see Presets.

CONTROL	WHAT IT DOES
Full rig	A capture with a cab in it. Parabola skips the IR for it.
Level matching	Every capture is brought to the same loudness, from its own data.

12 Pitch & Tuner *Drop-tune without touching the pegs, and tune silently.*

What it does. Pitch transposes your whole signal up or down, in semitones, before the bands, so a 4-string in E can play like it's in D or B. TUNER opens a tuner and mutes the output while it's open.

How it works. Pitch uses T3KJ's pitch shifter (a correlation-spliced delay line that keeps even low B clean and re-syncs on every pick attack). While Pitch isn't 0 it adds about 16 ms of delay, which Parabola reports to your DAW. The tuner listens to your DI before anything else.

The sound. -2 for drop D-style riffs on a standard-tuned bass, -5 for a B-standard feel. Bigger shifts get more artificial.

Try this.

1. Set Pitch to -2 and play an open E: you'll hear a D.
2. Click TUNER, play each open string, close it when they're green.

CONTROL	WHAT IT DOES
Pitch	-12 to +12 semitones. 0 is off (no delay).
TUNER	Note, cents and Hz; green when within 3 cents. Output muted while open.

13 MIDI & footswitches *Put any control on a footswitch or pedal.*

What it does. Every knob, power light, CAB switch and EQ fader can follow a MIDI CC from a controller (a Quad Cortex footswitch, an expression pedal).

How it works. Right-click a control, choose MIDI Learn, then press the footswitch or move the pedal. Switches follow the CC (64 and up is on). Mappings save with the plugin's state. In a DAW you can also use its own parameter learn. The standalone app needs your controller ticked under MIDI inputs in its audio settings.

The sound. Footswitch the high band on for choruses, or put the mid band's Drive on an expression pedal.

Try this.

1. Right-click the HIGH power light, MIDI Learn, press a footswitch.
2. Right-click it again to forget the CC.

CONTROL	WHAT IT DOES
Right-click	MIDI Learn / Forget / Cancel.
Hint bar	Shows a control's CC once it's mapped.

14 Hush *A de-noiser for your DI, with zero latency.*

What it does. Hush turns down the hiss and electrical buzz (EMI) your pickups bring, before anything in the chain amplifies it. It works between notes and also under them, in the parts of the spectrum your note isn't using. It's off until you turn it on: click HUSH in the header.

How it works. Hush watches eight bands from 200 Hz up. Each band has a noise floor (set by LEARN) and opens the moment your note shows up in it, within about two milliseconds, so pick attacks get through. A band with only noise in it is turned down with an EQ cut, by up to Strength. Nothing is delayed, and with every band open your signal is untouched. Below 200 Hz nothing changes: hum is a job for the gates (and, on a Jazz Bass, both pickup volumes full up).

The sound. A quiet floor under high gain: no hiss between riffs and less fizz around held notes, without a gate's chop. Too much Strength or Sensitivity can make quiet notes a little duller.



HUSH in the header opens it; DONE goes back. The bars show how much each band is being turned down.

Try this.

1. Click HUSH and turn it ON.
2. Volume up on your instrument, hands off the strings, click LEARN and wait two seconds.
3. Play, and raise Strength until the hiss is gone. If quiet notes lose sparkle, lower Sensitivity or Strength.
4. Use it with the gates, not instead of them: with Hush on you can set them lower, so notes fade out more naturally.

CONTROL	WHAT IT DOES
ON / OFF	Off is a perfect bypass.
LEARN	Two seconds of listening, hands off the strings, to set each band's noise floor. Saved with your settings.
Strength	How far a noise-only band is turned down (100% = 30 dB).
Sensitivity	How far over its floor a band has to be to count as your note (0 to 24 dB).
Release	How fast a band turns down once your note leaves it (10 to 500 ms).
Meters	How far each band is being turned down right now.

15 Hiss & noise *Why a high-gain band hisses, and what fixes it.*

What it does. A drive capture has a lot of gain, and it amplifies everything you feed it, including your bass's own noise: a passive single-coil (a Jazz Bass with one pickup down, say) hums and hisses at around -60 dB, and a high-gain capture can lift that to a loud fizz. The high band hears it most, because above the split a bass has little note to cover it.

How it works. Hush (header) is first in line: it learns your bass's noise and turns hiss down between notes and under them, with zero latency. Gate (header) works on your DI after it. Noise (header) is a second gate after the mid and high bands, following your DI: it shuts their hiss as each note fades, while the low band keeps ringing clean. Both gates use the same threshold idea: the level under which they close.

The sound. Silent stops between riffs, no fizz tail on held low notes, and a clean low end underneath.

Try this.

1. Jazz Bass: both pickup volumes all the way up (that's when they cancel each other's hum).
2. Click HUSH, turn it ON, hands off the strings, LEARN. Raise Strength until the fizz is gone.
3. Set Gate where the hiss between notes cuts out with your hands off the strings (often -55 to -45 dB; lower with Hush on).
4. Turn Noise up until the drive bands' fizz stops as notes fade (often -55 to -45 dB).
5. Still hissy while notes ring: lower the high band's Drive, bring its Tone down, or use a less extreme capture.

CONTROL	WHAT IT DOES
Hush	The de-noiser: see Hush.
Gate	DI gate, -90 (off) to -20 dB.
Noise	Drive bands' gate, -90 (off) to -20 dB.

16 Settings & latency *Skins, audio setup and delay.*

What it does. The gear opens Settings: five skins (Foil, Inferno, Hologram, Ram, Ember), and in the standalone app your audio interface, input, outputs, sample rate and buffer, with a latency readout.

How it works. Captures are trained at 48 kHz, so Parabola runs everything at 48 kHz. If your project or interface runs at another rate, it converts on the way in and out, which adds under 1 ms; it tells your DAW so tracks stay lined up. At 48 kHz it adds nothing.

The sound. For the lowest latency: ASIO, 48 kHz, and the smallest buffer that doesn't crackle (64 or 128 is typical). Three captures use more CPU than one; raise the buffer if it crackles.

Try this.

1. No sound in the standalone app? Turn on Hear yourself and check your bass's input is ticked.
2. Drag the bottom-right corner to resize the window (60% to 200%).

CONTROL	WHAT IT DOES
Skin	The photo and colors. Saved with the plugin's state.
Hear yourself	Plays the output while your input is live. Off by default to prevent feedback.